Co-Enhanced Global-Part Integration for Remote-Sensing Scene Classification

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***Abstract*— Remote-sensing (RS) scene classification aims to classify RS images with similar scene characteristics into one category. Plenty of RS images are complex in background, rich in content, and multiscale in target, exhibiting the characteristics of both intraclass separation and interclass convergence. Therefore, discriminative feature representations designed to highlight the differences between classes are the key to RS scene classification. Existing methods represent scene images by extracting either global context or discriminative part features from RS images. However, global-based methods often lack salient details in similar RS scenes, while part-based methods tend to ignore the relationships between local ground objects, thus weakening the discriminative feature representation. In this article, we propose to combine global context and part-level discriminative features within a unified framework called CGINet for accurate RS scene classification. To be specific, we develop a light context-aware attention block (LCAB) to explicitly model the global context to obtain larger receptive fields and contextual information. A co-enhanced loss module (CELM) is also devised to encourage the model to actively locate discriminative parts for feature enhancement. In particular, CELM is only used during training and not activated during inference, which introduces less computational cost. Benefiting from LCAB and CELM, our proposed CGINet improves the discriminability of features, thereby improving classification performance. Comprehensive experiments over four benchmark datasets show that the proposed method achieves consistent performance gains over state-of-the-art (SOTA) RS scene classification methods.**

***Index Terms*— Attention, convolutional neural networks (CNNs), discriminative part discovery, remote-sensing (RS), scene classification.**

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I. Introduction

As a vital role in the intelligent interpretation of remote-sensing (RS) images, RS scene classification aims to map the scene content in images into semantic labels. It can be applied in many real-life applications, such as national governance and security surveillance [1], [2]. However, RS scene classification still faces two main challenges: large intraclass diversity and large interclass similarity. In the first, intraclass diversity, scene images from the same class mostly contain the same representative ground objects. These ground objects can vary in color, size, and number. As depicted in Fig. 1(a), airplanes in different images show obvious differences in number and color. Moreover, the imaging results of RS scenes are easily affected by the external environment (such as weather and light). In the second, interclass similarity, RS scenes cover a wide geographic area, and scene images from different classes are very likely to exhibit similar geographical properties. Thus, there is a high degree of semantic overlap between some scene classes. As depicted in Fig. 1, both mobile home parks and medium-sized residential areas contain houses, and the houses take up almost the entire image. Therefore, improving the classification performance of scene images with complex backgrounds is still a desirable, albeit challenging, goal.

Recent years have witnessed the vast development of deep neural networks (DNNs) in the field of computer vision. DNNs have shown outstanding performance in nonlinear modeling to directly generate powerful feature representations in various vision tasks. Existing methods mostly employ DNNs to generate feature representations of RS scenes and have achieved significant performance improvements. Generally speaking, these methods can be divided into two categories: global-based methods and part-based methods. Global-based methods improve model performance by modeling the global context to

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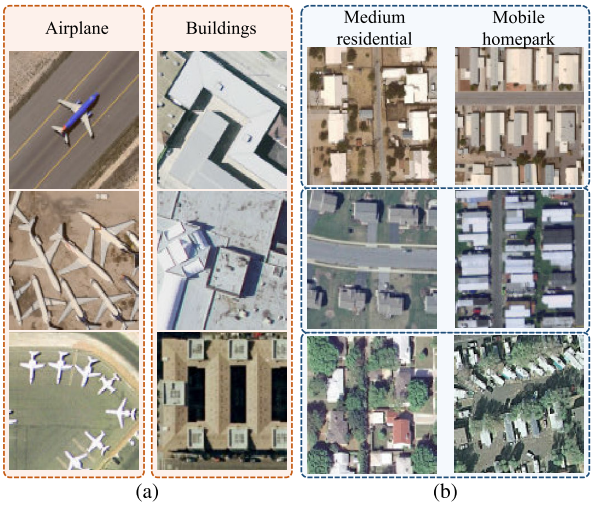


Fig. 1. RS scene images selected from the UCM dataset [3], demonstrate both (a) large intraclass diversity and (b) large interclass similarity. (a) RS scene images of aircraft and buildings that exhibit diverse image features. (b) RS scene images of mobile home parks and medium-sized residential areas that exhibit similar geographical properties with high semantic overlap.

alleviate local ambiguity. For example, the purpose of gated bidirectional network (GBNet) [4] is to integrate multilayer convolutional features to enhance the feature representation of scene images. Interestingly, experimental results on GBNet show that better performance is achieved by capturing global features. Part-based methods improve model performance by learning and localizing local discriminative parts of features. For example, the literature [5] cyclically integrates high-level semantic information and spatial information into the attention vector space and finally filters out key information. The recently developed multi-branch local attention network (MBLANet) [6] proposes a dual-attention structure to promote DNNs to perceive the discriminative area of input samples and ignore redundant information.

Nevertheless, the above-mentioned methods only focus on one aspect of features to represent scene images, while the rest are passively learned and insufficient. Specifically, global features lack salient details in similar scenes, while part features lack the global context of scenes and ignore the relationships between local ground objects. To address these limitations, an obvious approach is to redesign the structure of traditional DNNs. For instance, the literature [7] comprehensively considers the advantages of convolutional neural networks (CNNs) in local modeling and vision transformer (ViT) in global modeling and improves classification accuracy effectively by integrating multilayer local features and modeling global information. Despite all these improvements, methods for redesigning DNNs tend to increase the computational load, number of parameters, and inference time of the model.

In this article, we present a novel RS scene classification method called CGINet that integrates global context and part-level discriminative features within a unified framework. To this end, as shown in Fig. 2, we first develop a light context-aware attention block (LCAB) to explicitly model the global context and then devise a co-enhanced loss module (CELM) to capture part-level discriminative features. Specifically, LCAB introduces simplified self-attention into channel attention to obtain larger receptive fields and contextual information. It is appended on all residual blocks in the ResNet50 [8] architecture. In parallel, CELM encourages the model to actively locate discriminative parts for feature enhancement. In CELM, the feature channels are first grouped so that each group represents a class. Then, we directly constrain the feature channels based on two optimization objectives. One objective is to align scene classes and subfeature groups, thus ensuring that each subfeature group only locates the discriminative part features of the target class. The other is to enrich discriminative part features within each subfeature group, thus ensuring that as many local discriminative features as possible can be learned. Our proposed CELM is only used during training and not activated during inference, which introduces less computational cost. Furthermore, we propose a self-enhancing submodule to improve the spatial distribution of semantic subfeatures within the group. This enables us to further align class information and feature groups to better-fit scene images.

In conclusion, the contributions of this work are as follows.

1) We propose to combine global context and part-level discriminative features within a unified framework called CGINet for accurate RS scene classification.
2) We develop an LCAB to explicitly model the global context to obtain larger receptive fields and contextual information.
3) We design a CELM to guide the model to actively locate discriminative parts for feature enhancement.
4) Comprehensive experiments on four benchmark datasets confirm the effectiveness of the proposed CGINet for RS scene classification.

The rest of this work is organized as follows. Section II provides an overview of related approaches and loss functions widely used in classification. Section III elaborates on the structure of the proposed CGINet. Section IV shows the experimental process with comparative analysis. Section V summarizes this work.

II. Related Works

A. Remote-Sensing Scene Classification

Earlier works largely relied on hand-crafted feature descriptors, which are constructed by extracting visual attributes, such as color, texture, spectral, and shape [9], [10]. However, it is difficult for these methods to comprehensively describe the rich semantic information of RS images, which limits their performance in RS scene classification.

To enrich semantic information, middle-level feature-based methods encode and transform low-level local descriptors into global features, such as principal component analysis (PCA) and a bag of visual words (BoVW) [3], [11]. Naturally, these methods that extracted more global information quickly gained dominance when they were first introduced and refined. However, they inherently rely on hand-crafted features and lack flexibility in discovering highly complex structures.

DNNs have successfully broken the limitations of traditional hand-crafted features with their impressive extraction

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capabilities. However, due to the complex and variable spatial distribution in RS images, conventional DNNs have not achieved the expected high accuracy in RS scene classification. Much research is devoted to targeted improvements of popular DNNs. ResNet [8], GoogLeNet [12], and the ViT [13] are currently the most commonly used baselines. In particular, some methods aim to locate important regions containing discriminative cues. Attention recurrent convolutional network (ARCNet) [5] cyclically integrates high-level semantic information and spatial information into the attention vector space and finally filters out key information. To jointly encode spatial information and multilevel features, Zhang et al. [14] inputs the features extracted by CNNs into a newly designed capsule network (CapsNet) for reinforcement. Some methods aim to model global context from images to understand scenes globally. In [15], CapsNet are utilized to supplement the spatial relationships that CNN lacks, thereby enhancing feature representation. However, the time-consuming training problem faced by the CapsNet has always limited its further development in the RS community. Feature aggregation CNN (FACNN) [16] integrates feature learning and aggregation. Moreover, it uses label information to supervise the aggregation of intermediate features, helping to guide the efficient aggregation of features.

Both local discriminative features and global context are critical for DNNs to learn discriminative feature representation in scene images. SCViT [17] not only considers the local information between patches in ViT, but also integrates the global information in the spatial and channel domains of the image, ultimately significantly improving the performance of ViT. In [18], a method called key filter bank (KFB) is proposed to locate significant information. KFB can capture features that show class uniqueness while preserving global features. The global–local interactive learning mode proposed in [19] can effectively enrich the feature space. In this article, a novel method called CGINet is presented for RS scene classification, addressing this task by simultaneously modeling global context and discovering local discriminative parts. Notably, we only use the loss function during discriminative part discovery to guide the model to learn local discriminative features.

B. Classification Loss Functions

The choice of loss function is an essential aspect of deep learning. Cross-entropy loss (CE Loss) is a cornerstone that cannot be ignored in various visual classification tasks. It adds information entropy as prior information to classification problems and quantifies the difference between the predicted result and the true label. CE Loss is good at learning interclass information. However, it only cares about the predicted probability of the ground-truth label and ignores the difference with other negative classes, ultimately leading to the dispersion of learned features [20]. Center loss [21] learns the center of the deep features of each sample and uses these centers to reduce the intraclass differences. The large margin softmax (L-Softmax) [22] adds an angle constraint to the standard softmax loss to improve feature discrimination. Similar to the idea of L-Softmax, the angular softmax (A-Softmax) [23] further adds a weight constraint to achieve better performance. Based on the above work, several improvement methods with the same design concept but different structures have been proposed in succession [24], [25]. In addition, some methods focus on hard example mining. For example, focal loss [26] reduces the weight of simple samples in model training, thereby quickly focusing on difficult samples that are helpful for model training.

Unlike the above loss functions, mutual-channel (MC) loss [27] exhibits impressive performance by explicitly encouraging DNNs to capture local discriminative features. It reduces the confusion between features and the difficulty of optimization by combining feature channels with class relations. Inspired by MC loss, channel correlation (CC) loss [28] utilizes a channel attention module to generate corresponding channel attention for each sample and then constrains these attention vectors. On the basis of these approaches, we propose a novel CELM, which aims to supervise the model to locate and learn local discriminative regions of features.

III. Methodology

The framework of CGINet is displayed in Fig. 2. CGINet integrates global context and part-level discriminative features within a unified framework to extract more discriminative features. First, an LCAB is developed to explicitly model the global context. Specifically, LCAB introduces simplified self-attention into channel attention. After LCAB aggregates the features of all positions to gather global contextual information, channelwise interdependencies are captured and finally enhanced features are obtained. LCAB is then appended on all residual blocks in the ResNet50 [8] architecture. In parallel with LCAB, a CELM is devised to capture part-level discriminative features. More details are provided in Sections III-A and III-B.

A. Light Context-Aware Attention Block

Global context plays a considerable role in DNNs' understanding of scene images with global receptive fields. The convolution operation builds pixel relationships on the local receptive field based on convolution filter weights. The global context is captured by stacking convolutional layers deeply. However, directly stacking more layers may lead to increased training error and impair model performance. In this article, we propose an LCAB, which develops simplified self-attention to model the global context explicitly and then recalibrates the weights of each channel.

The structure of LCAB is illustrated in Fig. 2. LCAB takes a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ as input, where C, H, and W represent the number of channels, spatial height, and width, respectively. First, a 1×1 convolution layer is applied to aggregate the feature map $\mathbf{F}$ across the channel dimension. Subsequently, a softmax layer is introduced to derive a spatial descriptor $\boldsymbol{v} \in \mathbb{R}^{1\times H\times W}$. The shapes of $\mathbf{F}$ and $\boldsymbol{v}$ are then adjusted to $\mathbf{F}' \in \mathbb{R}^{C\times 1\times HW}$ and $\boldsymbol{v}' \in \mathbb{R}^{HW\times 1}$. A matrix multiplication is conducted between $\mathbf{F}'$ and $\boldsymbol{v}'$ to yield a $C\times 1\times 1$ channel descriptor. This process can be interpreted as an attention-pooling operation for adaptively modeling global

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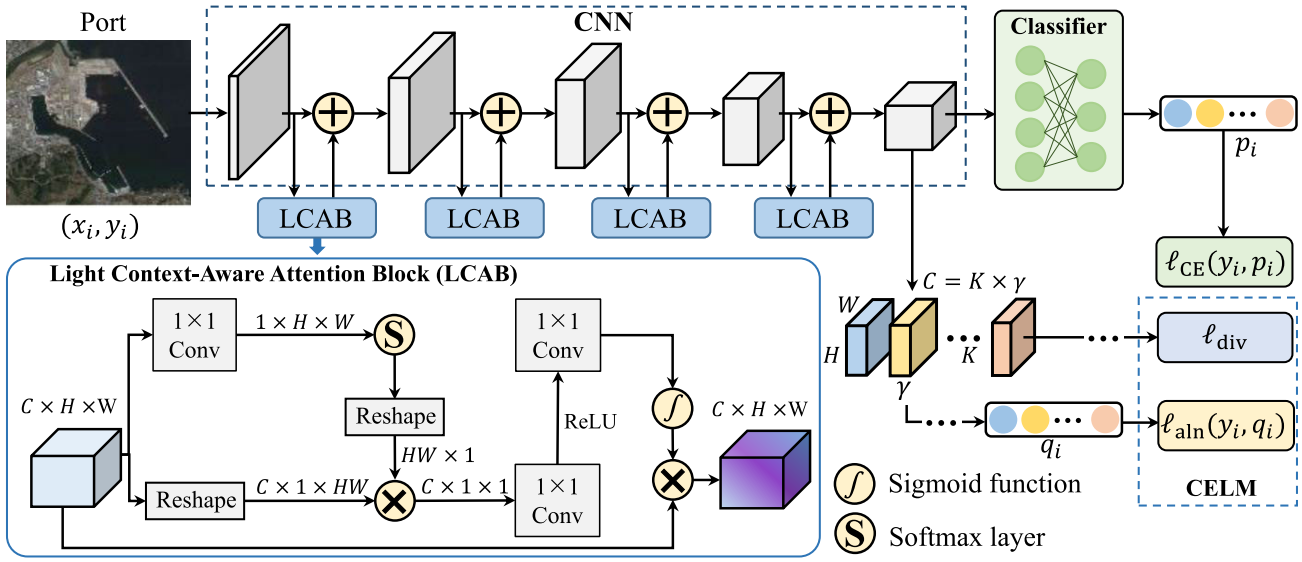


Fig. 2. Overall architecture of CGINet. It contains two main parts: an LCAB and a CELM. LCAB introduces simplified self-attention into channel attention. By first aggregating the features of all positions to gather global information, then capturing channelwise interdependencies, and finally outputting enhanced features. LCAB is then appended on all residual blocks in the convolutional backbone network. The widely used ResNet50 [8] is adopted as the convolutional backbone network of CGINet. In parallel, CELM supervises the model to locate and learn local discriminative regions of features.

context. Following this, two layers of 1×1 convolution are employed to capture the dependencies between different channels, and the output size of the first layer is set to $\mathbb{R}^{C/r \times 1 \times 1}$, where r is the reduction ratio. An attention map $\mathbf{F}_c \in \mathbb{R}^{C \times 1 \times 1}$ is subsequently generated using the sigmoid activation function. Finally, a matrix multiplication between $\mathbf{F}_c$ and $\mathbf{F}$ is performed to obtain the enhanced feature

$$\mathbf{F}_{\text{final}} = \mathbf{F} \cdot \mathbf{F}_c \tag{1}$$

with

$$\mathbf{F}_c = \sigma\left(f^{1\times1}\left(f^{1\times1}\left(\mathbf{F}' \cdot \boldsymbol{v}'\right)\right)\right) \tag{2}$$

the term $\mathbf{F}' \cdot \boldsymbol{v}'$ denotes the process of modeling global context, where pixels at all locations are pooled together by a weighted average of $\boldsymbol{v}'$. The term $f^{1\times1}$ indicates a 1×1 convolution layer, while σ denotes the sigmoid activation function.

B. Co-Enhanced Loss Module

Most previous methods resort to additional model structures for learning discriminative part features [5], [14]. However, they are prone to problems with slower convergence and redundant model parameters. In DNNs, features consist of multiple channels, with each channel representing part of the image information. Ideally, each channel has the potential to perceive part-level features of the image [27]. We group the features extracted by ResNet50 evenly based on the classes of the dataset, and average each subfeature group before displaying it. It can be seen from Fig. 3 that the discriminative part features learned are mixed together. To this end, we are inspired to devise a novel CELM to guide the model to actively locate discriminative parts for feature enhancement. An important characteristic of CELM is to enforce RS scene class constraints on the feature channels. During training, the LCAB-based backbone network and classifier can be optimized in an end-to-end manner thanks to CELM. During inference, we only need to input the features extracted by the LCAB-based backbone network into the classifier to predict the most likely class.

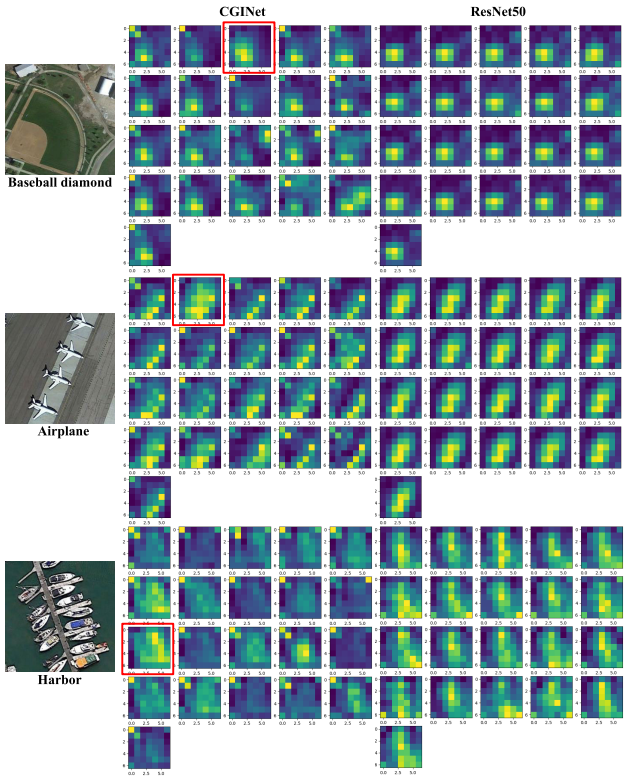


Fig. 3. Taking the UCM dataset as an example, the features within each subfeature group are averaged and displayed. Among the features extracted by CGINet, it can be found that the subfeature group of the target class has a stronger semantic response. Note that the red boxes mark the subfeature group of the target class of the sample.

1) Preliminary: Given a training set $\{x_i, y_i\}_{i=1}^{N}$ with K classes, where x_i and y_i refer to the input image and its semantic label, respectively, we use the LCAB-based backbone

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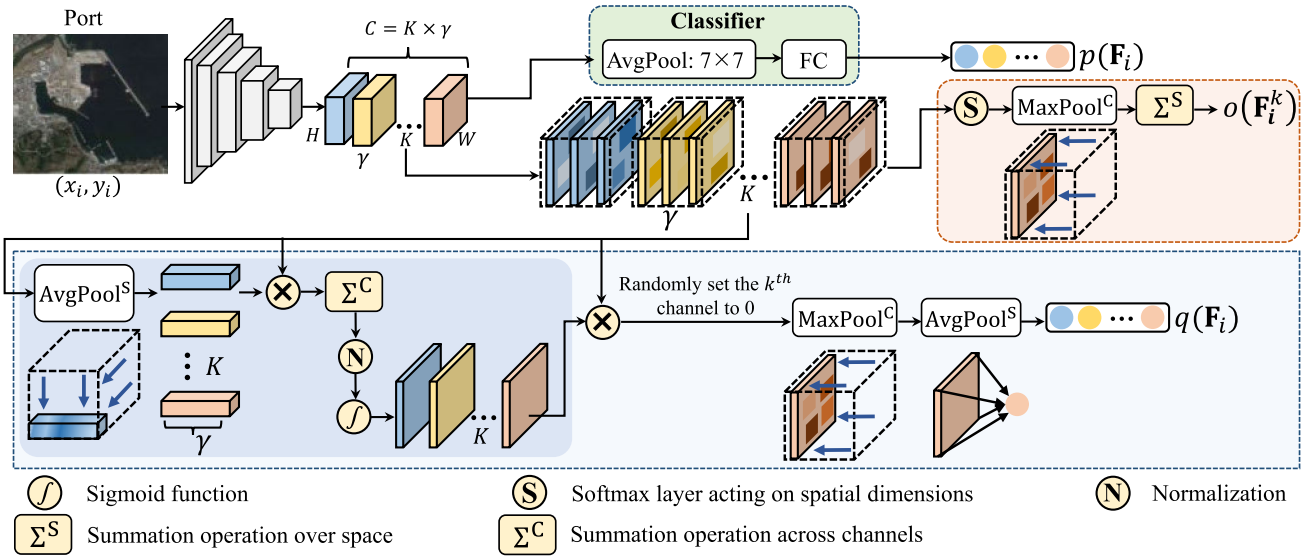


Fig. 4. Proposed CELM. First, the feature maps are uniformly grouped along the channel based on the number (K) of classes, that is, $C = K \times \gamma$, where γ denotes the number of channels of subfeature groups. Then, the feature $\mathbf{F}_i$ is fed into two branches, and loss values for two different optimization objectives are calculated. One branch aligns scene classes and subfeature groups (i.e., ℓ_{aln}), thus ensuring that each subfeature group only locates the discriminative part features of the target class. The other enriches discriminative part features within each subfeature group (i.e., ℓ_{div}), thus ensuring that as many local discriminative features as possible can be learned.

network to generate the feature maps. For each feature map $\mathbf{F}$ where $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the classifier employs a softmax layer to output the predicted probability $p(k|\mathbf{F})$ of class k. The standard CE Loss function used in RS scene classification can be defined as

$$\ell_{\text{CE}} = -\sum_{k=0}^{K-1} \mathbb{I}(k = y_i)\log(p(k|\mathbf{F})) \tag{3}$$

where $\mathbb{I}() \in \{0, 1\}$ is an indicator function. The value is 1 when the element satisfies the input condition.

2) Feature Grouping: CELM first divides the feature map $\mathbf{F}$ into K groups along the channel dimension, that is, $\mathbf{F} = [\mathbf{F}(0), \ldots, \mathbf{F}(K-1)]$, $\mathbf{F}(k) \in \mathbb{R}^{\gamma\times H\times W}$, where $C = K \times \gamma$. Here, K and γ represent the number of classes and channels in each subfeature, respectively. During training, each subfeature group $\mathbf{F}(k)$ gradually captures a specific semantic response.

3) Intergroup Alignment and Within-Group Diversity: Next, we fed $\mathbf{F}$ into two branches and calculated loss values for two optimization objectives. One objective is to align scene classes and subfeature groups, thus ensuring that each subfeature group focuses on the discriminative part features of the target class. The other is to enrich discriminative part features within each subfeature group, thus ensuring that as many local discriminative features as possible can be learned. As shown in Fig. 4, CELM comprises two losses: an intergroup alignment loss (ℓ_{aln}) to align scene classes and subfeature groups, and a within-group diversity loss (ℓ_{div}) to enrich discriminative part features within each subfeature group.

a) Intergroup alignment loss: RS scene images are diverse in geographical features, complex in backgrounds, and multiscale in target. It is difficult to obtain representative class-level statistics simply by compressing feature maps in spatial and channel dimensions. To better align scene classes and subfeature groups, we first design a self-enhancing submodule to enhance the semantic regions of features. The self-enhancing submodule explicitly models the correlation of global and local features in subfeature groups. More specifically, the global semantic information is approximated by the global statistical features, which are directly multiplied by the local features within the group

$$\mathbf{g}(k) = \text{AvgPool}^{\text{s}}(\mathbf{F}(k)) = \frac{1}{HW}\sum_{j=1}^{HW} \mathbf{F}_j(k) \tag{4}$$

$$\mathbf{e}(k) = \mathbf{F}(k) \cdot \mathbf{g}(k) \tag{5}$$

where $\text{AvgPool}^{\text{s}}$ denotes the global average pooling over the space. Note that $\mathbf{e}(k)$ can also be expressed as $\mathbf{e}(k) = \|\mathbf{F}(k)\|\|\mathbf{g}(k)\|\cos(\theta_k)$, where θ_k is the angle between $\mathbf{F}(k)$ and $\mathbf{g}(k)$. Generally speaking, different samples often have different feature scales. To prevent the biased magnitude of coefficients among different samples, we normalize $\mathbf{e}(k)$ over space to obtain $\hat{\mathbf{e}}(k)$, as is widely used in [29] and [30]. Then, $\hat{\mathbf{e}}(k)$ is scaled by the sigmoid function $\sigma()$ over the space. The subfeature group enhanced by the self-enhancing submodule can be expressed as

$$\hat{\mathbf{F}}(k) = \mathbf{F}(k) \cdot \sigma\left(\hat{\mathbf{e}}(k)\right). \tag{6}$$

Based on the enhanced subfeature groups, the intergroup alignment loss ℓ_{aln} for $\mathbf{F}$ is formulated as

$$\ell_{\text{aln}} = -\sum_{k=0}^{K-1} \mathbb{I}(k = y_i)\log\left(q\left(k|\hat{\mathbf{F}}\right)\right) \tag{7}$$

with

$$q\left(k|\hat{\mathbf{F}}\right) = \text{AvgPool}^{\text{s}}\left(\text{MaxPool}^{\text{c}}\left(\Theta\left(\hat{\mathbf{F}}(k)\right)\right)\right) \tag{8}$$

where $\mathbb{I}()$ is an indicator function in the range $\{0, 1\}$. The value is 1 when the element satisfies the input condition. Θ denotes a random channel discard operation with a probability of 0.5. In this way, each channel feature is actively promoted to contain as much discriminative information as possible.

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$\text{MaxPool}^{\text{c}}$ and $\text{AvgPool}^{\text{s}}$ represent the global max pooling across channels and the global average pooling over the space, respectively.

b) Within-group diversity loss: RS scenes have larger interclass similarities, so it is necessary to facilitate the model to learn the feature diversity within each scene class. As shown in [27], feature diversity can be reflected by the different spatial distributions of features, and different spatial distributions display different semantic information. Fig. 3 can intuitively reflect that feature diversity can be represented by its spatial semantic distribution. To this end, the within-group diversity loss ℓ_{div} for $\mathbf{F}$ is formulated as

$$\ell_{\text{div}} = -\frac{1}{K \times \gamma} \sum_{k=0}^{K-1} o(\mathbf{F}(k)) \tag{9}$$

with

$$o(\mathbf{F}(k)) = \Sigma^{\text{s}}\left(\text{MaxPool}^{\text{c}}(\text{softmax}(\mathbf{F}(k)))\right) \tag{10}$$

where softmax() represents the softmax function, which is used to normalize each feature over space. $\text{MaxPool}^{\text{c}}$ denotes the max pooling operation across channels, and Σ^{s} denotes the summation operation over space.

4) Overall Loss: Finally, we use a novel loss module (CELM) to guide the model to locate and learn local discriminative regions of features. The overall loss function is defined as

$$\mathcal{L} = \ell_{\text{CE}} + \beta \times (\ell_{\text{aln}} + \eta \times \ell_{\text{div}}) \tag{11}$$

where β is used to balance CE loss and CELM, and η is a coefficient to tradeoff ℓ_{aln} and ℓ_{div}.

IV. Experiments

A. Data Description

To confirm the effectiveness of CGINet, four public datasets are selected: the UC Merced (UCM) dataset [3], the AID dataset [2], the NWPU-RESISC45 (NWPU) dataset [31], and the OPTIMAL-31 dataset [5].

1) The UCM dataset [3] consists of 21 RS scene categories, where each class contains 100 images with a resolution of 256 × 256. It is collected from the USGS national map and is the most widely used dataset in the RS community.
2) The AID dataset [2] consists of 10 000 images divided into 30 scene classes, where each class contains at least 200 images with a resolution of 600 × 600 pixels.
3) The NWPU dataset [31] consists of 45 RS scene categories. Each class contains 700 images with a resolution of 256 × 256 pixels. The NWPU dataset is currently one of the largest datasets in the RS community, with a total of 31 500 images. The entire dataset is characterized by a wide distribution of collection areas and variable imaging conditions. The resolution range is from 0.2 to 30 m.
4) The OPTIMAL-31 [5] dataset consists of 31 RS scenes, and each class includes 60 images extracted from Google Earth with a resolution of 256 × 256 pixels.

B. Implementation Details

The model framework is implemented on the PyTorch and MindSpore framework, and a GTX2080Ti GPU is used to conduct all experiments. We adopt the widely used ResNet50 [8] as the backbone network of CGINet. Random horizontal flips and random rotation are used to augment the data during training. All input images are resized into 224×224 pixels. The number of iterations during training is 100 and the batch size is set to 32. The stochastic gradient descent (SGD) algorithm with momentum set to 0.9 is selected as the optimizer. In addition, the learning rate is initialized to 0.01, and a cosine decay strategy is adopted. The backbone network is initialized with the weight parameters trained in ImageNet [32].

All images of the dataset are randomly assigned to either the training set or the test set. For a fair comparison, these datasets are divided in the same way as [6] and [33]. Specifically, the proportion of training data in UCM and OPTIMAL-31 is 80%. The proportions of training data in AID are 20% and 50%, respectively. The proportions of training data in NWPU are 10% and 20%, respectively.

The reduction ratio r in LCAB is set to 16. The number of feature channels output by the pretrained ResNet50 is 2048, and the number of classes for different datasets is different. Therefore, it is not necessarily possible to make the subfeature groups of each class have the same number of channels. Referring to [27], we distribute γ nonuniformly to use pretrained ResNet50. Specifically, for a dataset with K classes and the number of channels output by the backbone network denoted as C, each of the top $K - C\%K$ classes is assigned $C//K$ feature channels. Each of the remaining $C\%K$ classes is assigned $C//K + 1$ feature channels. The hyperparameter terms β and η are set as follows. β is set to 0.3 and η is set to 5 in the UCM dataset. When the training ratio in AID is 20%, β is set to 0.3 and η is set to 5. When the training ratio in AID is 50%, β is set to 0.2 and η is set to 5. In the NWPU dataset, β is set to 0.1 and η is set to 5. In the OPTIMAL-31 dataset, β is set to 0.3 and η is set to 6. The values of the above hyperparameters β and η are all set based on experimental experience to achieve better results in the corresponding datasets.

Overall accuracy (OA) is used as an indicator to evaluate the effectiveness of the method, which reflects the overall classification performance of the model. In addition, to intuitively reflect the degree of classification errors and confusion between classes, the confusion matrix (CM) and Kappa coefficient (Kappa) are selected as the other two indicators. All experiments are repeated five times and the average results and standard deviations are reported to avoid any bias caused by random sampling.

C. Comparison With State-of-the-Art Methods

To verify the effectiveness of CGINet, some state-of-the-art (SOTA) methods are applied for comparison on four public benchmark datasets. According to the type of backbone network employed by these methods, we classify these methods into four categories: CNN-based methods, ViT-based methods, CNN-ViT-based methods, and multilayer perceptron (MLP)-based methods. In particular, given that the backbone network of CGINet is CNN, more CNN-based methods

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TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON UCM

Methods	Backbone	80% training
GooLeNet [12]	GooLeNet	94.31±0.89
VGGNet16 [34]	VGG	95.21±1.20
ResNet50+CBAM [35]	ResNet	99.28±0.24
SCCov [37]	VGG	99.05±0.25
VGG-16-CapsNet [14]	VGG	98.81±0.22
ARCNet-VGGNet16 [5]	VGG	99.12±0.40
MG-CAP (Sqrt-E) [38]	VGG	99.00±0.10
GBNet+global feature [4]	VGG	98.57±0.48
HHTL [44]	ViT	99.48±0.28
DFAGCN [39]	GCN	98.48±0.42
ResNet50+EAM [36]	ResNet	98.98±0.37
ACR-MLFF [40]	ResNet	99.37±0.15
T2T-ViT-12 [45]	ViT	99.10±0.32
ACNet [41]	VGG	99.76±0.10
EFPN-DSE-TDFF [42]	ResNet	99.14±0.22
ViT-B [13]	ViT	99.28±0.23
PVT-V2-B0 [46]	ViT	98.86±0.38
Cycle MLP [47]	MLP	99.05±0.39
EMTCAL [7]	ViT+ResNet	99.57±0.28
MBLANet [6]	ResNet	99.52±0.23
EMSCNet (ResNet50) [43]	ResNet	99.44±0.16
ResNet50 [8]		99.22±0.34
CGINet (ours)		**99.84±0.16**

Note: 80% of the data is used for training.

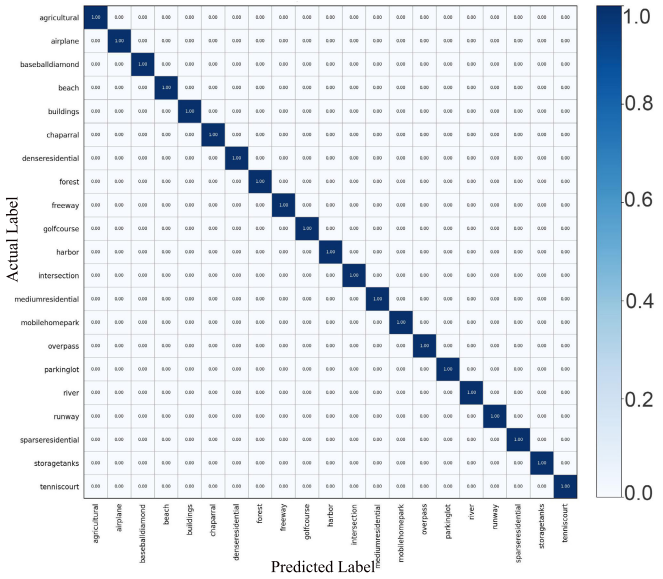


Fig. 5. CM on UCM when the proportion of trainable data is 80% and OA equals 100.00%.

are presented in the comparison than other categories of methods.

1) Comparison on UCM: Table I reports the OA values of CGINet and other methods on UCM. The results show that the following holds.

1) The proposed CGINet can achieve 99.84%, surpassing all compared methods. The positive results demonstrate the effectiveness of CGINet on the small-scale dataset.
2) Among the classic CNNs (e.g., GoogLeNet [12], VGGNet16 [34], and ResNet50 [8]), ResNet50 achieves the best result. ResNet relies on convolution operations with local receptive fields to extract features and achieves a basic performance of 99.22%. It is worth noting that even if an attention mechanism (e.g., convolutional block attention module (CBAM) [35] and efficient attention module (EAM) [36]) that can capture global information is introduced, the improvement of OA is not obvious. This suggests that effective mechanisms that work for natural images may not necessarily work for the more complex RS images. Based on this observation, the proposed CGINet aims to capture global and local discriminative features simultaneously, ultimately improving OA results by 0.62 percentage points.
3) When an attention mechanism designed for the characteristics of RS images is introduced, the baseline can achieve higher accuracy. MBLANet [6] embeds an attention-based feature refinement module [convolutional local attention module (CLAM)] into ResNet50 to better capture key information from redundant information. Despite its effectiveness, our method still improves by 0.32 percentage points over MBLANet. This suggests that better results can be obtained by actively localizing local discriminative regions.
4) In comparison with these CNN variants, the percentage point improvements obtained by CGINet are 0.79 (over skip-connected covariance network (SCCov) [37]), 1.03 (over VGG-16-CapsNet [14]), 0.72 (over ARCNet-VGGNet16 [5]), 0.84 (over multi-granularity canonical appearance pooling (MG-CAP) (Sqrt-E) [38]), 1.27 (over GBNet+global feature [4]), 1.36 (over deep feature aggregation framework driven by graph convolutional network (DFAGCN) [39]), 0.86 (over ResNet50+EAM [36]), 0.47 (over multi-level feature fusion networks with adaptive channel dimensionality reduction (ACR-MLFF) [40]), 0.08 (over attention consistent network (ACNet) [41]), 0.7 (over enhanced feature pyramid network with deep semantic embedding and two-branch deep feature fusion (EFPN-DSE-TDFF) [42]), 0.32 (over MBLANet [6]), and 0.4 (over efficient multisample contrastive network (EMSCNet) (ResNet50) [43]). These encouraging results demonstrate the positive role of CGINet in improving the performance of baseline models for RS scene classification.

The CM of CGINet on UCM, when the training data accounts for 80% of the dataset, is depicted in Fig. 5. It can be clearly observed that CGINet has no confusion for each category of UCM. The Kappa value of CGINet is high (0.9962). The positive results indicate that the proposed CGINet is absolutely competitive on the small-scale but widely used UCM dataset.

2) Comparison on AID: Table II reports the OA values of CGINet and other methods on the AID dataset. The results show that the following holds.

1) The proposed CGINet achieves the best results among the compared methods. ViT-based methods have obvious advantages over CNNs in modeling global context due to their unique self-attention mechanism. Therefore,

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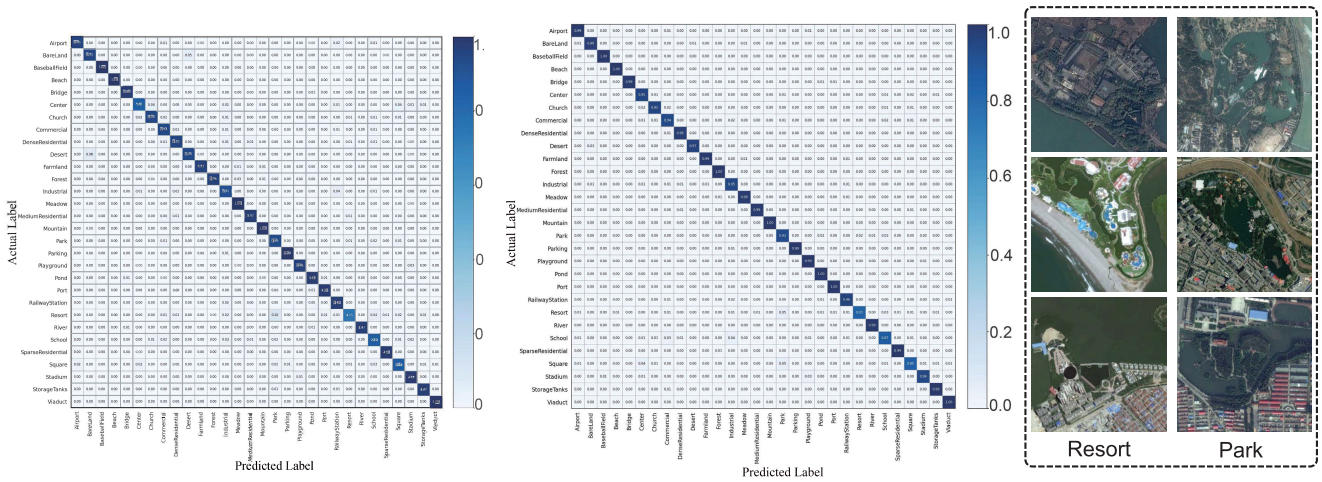


Fig. 6. CM and image examples for AID with 20% and 50% trainable data. The OA results are 95.48% and 97.10%, respectively. (a) CM (training data accounts for 20%). (b) CM (training data accounts for 50%). (c) Image examples of the AID dataset.

TABLE II
EXPERIMENTAL RESULT OF DIFFERENT METHODS ON AID

Methods	Backbone	OA (%)	
		20% Training	50% Training
GooLeNet [12]	GooLeNet	83.44±0.40	86.39±0.55
VGGNet16 [34]	VGG	86.59±0.29	89.64±0.36
ResNet50+CBAM [35]	ResNet	94.46±0.14	96.41±0.38
SCCov [37]	VGG	93.12±0.25	96.10±0.16
VGG-16-CapsNet [14]	VGG	91.63±0.19	94.74±0.17
ARCNet-VGGNet16 [5]	VGG	88.75±0.40	93.10±0.55
GBNet+global feature [4]	VGG	92.20±0.23	95.48±0.12
MG-CAP (Sqrt-E) [38]	VGG	93.34±0.18	96.12±0.12
DFAGCN [39]	GCN	-	94.88±0.22
ACR-MLFF [40]	ResNet	92.73±0.12	95.06±0.33
T2T-ViT-12 [45]	ViT	94.39±0.22	96.29±0.24
ACNet [41]	VGG	93.33±0.29	95.38±0.29
EFPN-DSE-TDFF [42]	ResNet	94.02±0.21	94.50±0.30
ViT-B [13]	ViT	93.81±0.21	96.08±0.14
ResNet50+EAM [36]	ResNet	94.26±0.11	97.06±0.19
PVT-V2-B0 [46]	ViT	93.52±0.35	96.27±0.14
Cycle MLP [47]	MLP	95.31±0.15	96.87±0.17
EMTCAL [7]	ViT+ResNet	94.69±0.14	96.41±0.23
MBLANet [6]	ResNet	94.62±0.19	96.56±0.24
EMSCNet (ResNet50) [43]	ResNet	95.13±0.10	96.96±0.10
ResNet50 [8]		94.47±0.18	96.20±0.24
CGINet (ours)		**95.35±0.14**	**97.10±0.24**

Note: 20% and 50% of the data is used for training.

we propose LCAB to develop simplified self-attention to model the global context explicitly, thereby compensating for the shortcomings of CNN in obtaining global information. Meanwhile, with the help of CELM, our method generates SOTA results with OA values of 95.35% and 97.10%, respectively. Better experimental results demonstrate the effectiveness of our CGINet.

2) Cycle MLP is built upon the cycle fully connected layer, which can handle variable input scales. Its comparable results with CGINet show the promise of MLP-based methods in the RS community. However, the training difficulties caused by its excessive parameter size are still urgent problems that need to be solved. Compared with MLP and ViT, CNN has been used longest and more commonly in RS scene classification. Therefore, we still consider CNN to be our backbone network.

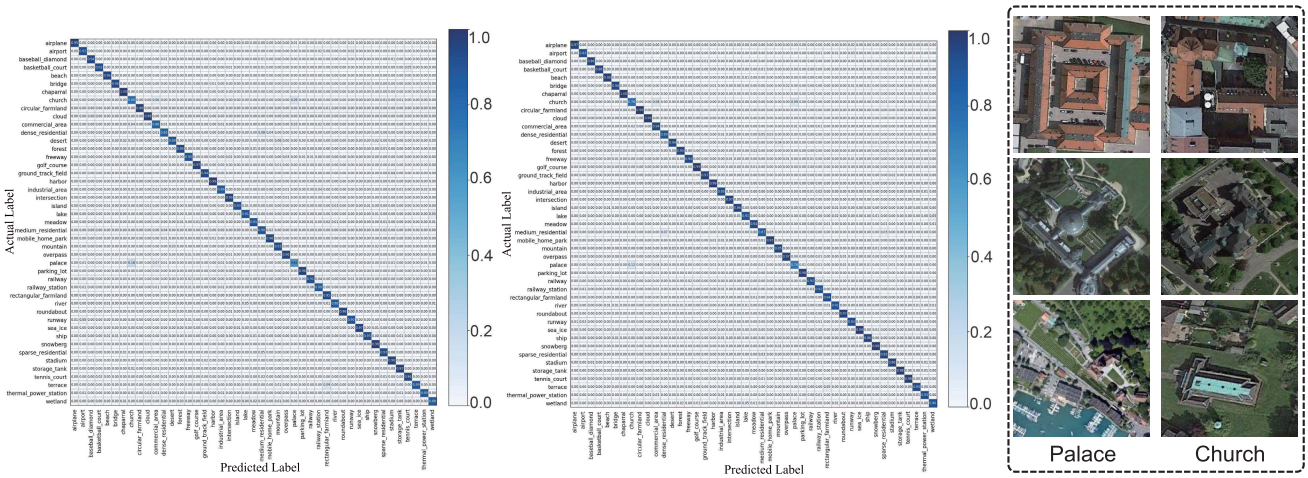
3) The OA values of classic ViT are 93.81% and 96.08%, achieving comparable results to those CNN-based methods. Although the performance of ViT-based methods (e.g., T2T-ViT-12 [45] and PVT-V2-B0 [46]) has improved compared to the original ViT, it is still slightly inferior to some optimized CNN-based methods. Generally speaking, ViT-based methods pay more attention to global features due to their inherent self-attention structure. However, local discriminative features are also important, since most RS images are of small scale. Experimental results show that the proposed CGINet is more conducive to improving classification performance by focusing on local features and global context simultaneously.

Apart from OA, Kappa values of CGINet are equal to 0.9510 and 0.9676 when the training data ratio is 20% and 50%, respectively. The CMs of CGINet, when 20% and 50% of the dataset are used for training, are reported in Fig. 6. It can be seen that CGINet can correctly distinguish most scene categories. In addition, under different training ratios, the classification accuracy exceeds 90% for 26 of the 30 categories. The above experimental results indicate that CGINet can clearly distinguish most categories, except for "Resort." By observing the "Resort" sample, we find that this category contains dense buildings and complex surface features. As shown in Fig. 6(c), the "Resort" category is very similar to the "Park" category.

3) Comparison on NWPU: NWPU is the largest dataset in our experiments. Table III reports the OA values of CGINet and other methods on NWPU. The results show that the following holds.

1) The proposed CGINet achieves the best performance with an accuracy of 92.28% and 94.38%, respectively, under the 10% and 20% training ratios, which substantially improves the accuracy by large margins. CGINet achieves consistent improvements over other methods. Specifically, the percentage point improvements in OA achieved by CGINet when the trainable data accounts for 10% are 0.12 (over EMSCNet (ResNet50) [43]), 0.65 (over efficient multiscale transformer and cross-level

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(a) (b) (c)

Fig. 7. CM and image examples for NWPU with 10% and 20% trainable data. The OA results are 92.32% and 94.38%, respectively. (a) CM (training data accounts for 10%). (b) CM (training data accounts for 20%). (c) Image examples of the NWPU dataset.

TABLE III
EXPERIMENTAL RESULT OF DIFFERENT METHODS ON NWPU

Methods	Backbone	OA (%)	
		10% Training	20% Training
GooLeNet [12]	GooLeNet	76.19±0.38	78.48±0.26
VGGNet16 [34]	VGG	76.47±0.18	79.79±0.15
ResNet50+CBAM [35]	ResNet	91.74±0.12	94.11±0.08
SCCov [37]	VGG	89.30±0.35	92.10±0.25
VGG-16-CapsNet [14]	VGG	85.08±0.13	89.18±0.14
MG-CAP (Sqrt-E) [38]	VGG	90.83±0.12	92.95±0.11
DFAGCN [39]	GCN	-	89.29±0.28
ACR-MLFF [40]	ResNet	90.01±0.33	92.45±0.20
T2T-ViT-12 [45]	ViT	90.62±0.18	93.19±0.10
ACNet [41]	VGG	91.09±0.13	92.42±0.16
ViT-B [13]	ViT	90.96±0.08	93.36±0.17
ResNet50+EAM [36]	ResNet	91.91±0.22	94.29±0.09
PVT-V2-B0 [46]	ViT	89.72±0.16	92.95±0.09
Cycle MLP [47]	MLP	91.84±0.17	94.27±0.11
EMTCAL [7]	ViT+ResNet	91.63±0.19	93.65±0.12
MBLANet [6]	ResNet	91.93±0.20	94.33±0.01
EMSCNet (ResNet50) [43]	ResNet	92.16±0.07	94.08±0.20
ResNet50 [8]		91.69±0.13	93.79±0.11
CGINet (ours)		**92.28±0.17**	**94.38±0.13**

Note: 10% and 20% of the data is used for training.

attention learning (EMTCAL) [7]), 0.44 (over Cycle MLP [47]), and 0.35 (over MBLANet [6]). Moreover, the percentage point improvements in OA achieved by CGINet when the trainable data accounts for 20% are 0.3 (over EMSCNet (ResNet50) [43]), 0.05 (over MBLANet [6]), 0.11 (over Cycle MLP [47]), and 1.96 (over ACNet [41]). These encouraging results confirm the superiority of the proposed CGINet on large-scale datasets.

2) Compared with our baseline (i.e., ResNet50), CGINet achieves 0.59 and 0.59 percentage point improvements, respectively. We believe that the introduction of LCAB and CELM enhances the baseline's feature learning ability in RS scenes with complex backgrounds.
3) Compared with CNN-based methods, CGINet improved by 0.12–16.09 percentage points. Compared with ViT-based methods, CGINet improved by 1.32–2.56 percentage points. Notably, the recently proposed EMTCAL [7] combines the advantages of the CNN in bias induction and ViT in effectively capturing long-distance dependencies to mine clues hidden in RS scene images. Nonetheless, our method still achieves 0.65 and 0.72 percentage point improvements, respectively.

The CMs of CGINet under different proportions of trainable data are presented in Fig. 7. When the proportion of trainable data is 10% and 20%, the model's accuracy for 34 and 41 categories, respectively, out of the 45 categories exceeds 90%. However, CGINet does not meet our expectations in some scenes, such as "church" and "Palace." Further analysis finds that these confusing scenes have a high interclass similarity. Solving this limitation will be a future direction for improving CGINet. Kappa values of CGINet are equal to 0.9193 and 0.9420 when the training data ratio is 10% and 20%, respectively.

4) Comparison on OPTIMAL-31: Table IV reports the OA values of CGINet and other methods on OPTIMAL-31. The results show that the following holds.

1) The proposed CGINet achieves the highest classification accuracy of 97.35%. Notably, we achieved an absolute improvement of up to 1.39 percentage points compared to the baseline ResNet50.
2) Recent CNN-based methods rely on attention mechanisms or feature aggregation strategies (e.g., MBLANet [6], GBNet [4], and attention cascade global–local network (ACGLNet) [33]). The optimization difficulty of these methods will increase as additional networks. In addition, the auxiliary network structure is not discarded during inference, which slows down the inference speed. In contrast, the additional parameters added by our proposed LCAB and CELM are not obvious. At the same time, CELM is only activated during model training without affecting the model inference rate, which is more efficient.
3) CGINet achieves consistent improvements compared to other methods. Quantitatively, the percentage point improvements in OA obtained by CGINet are 10.2 (over GooLeNet [12]), 2.48 (over DenseNet121 [48]), 1.92 (over ResNet50+CBAM [35]), 4.65 (over ARCNet-VGGNet16 [5]), 0.9 (over ResNet50+EAM [36]), 4.07 (over GBNet+global feature [4]), 2.23

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TABLE IV
EXPERIMENTAL RESULT OF DIFFERENT METHODS ON OPTIMAL-31

Methods	Backbone	80% Training
GooLeNet [12]	GooLeNet	87.15±0.45
DenseNet121 [48]	DenseNet	94.87±0.32
ResNet50+CBAM [35]	ResNet	95.43±0.27
ARCNet-VGGNet16 [5]	VGG	92.70±0.35
ResNet50+EAM [36]	ResNet	96.45±0.28
GBNet+global feature [4]	VGG	93.28±0.27
KFBNet(VGGNet-16) [18]	VGG	95.12±0.13
KFBNet(DenseNet-121) [18]	DenseNet	95.60±0.63
ViT-B [13]	ViT	95.97±0.18
ACGLNet [33]	VGG	96.02±0.24
VSDNet-DenseNet121 [49]	DenseNet	96.95±0.17
MBLANet [6]	ResNet	96.50±0.25
ResNet50 [8]		95.96±0.23
CGINet (ours)		**97.35±0.14**

Note: 80% of the data is used for training.

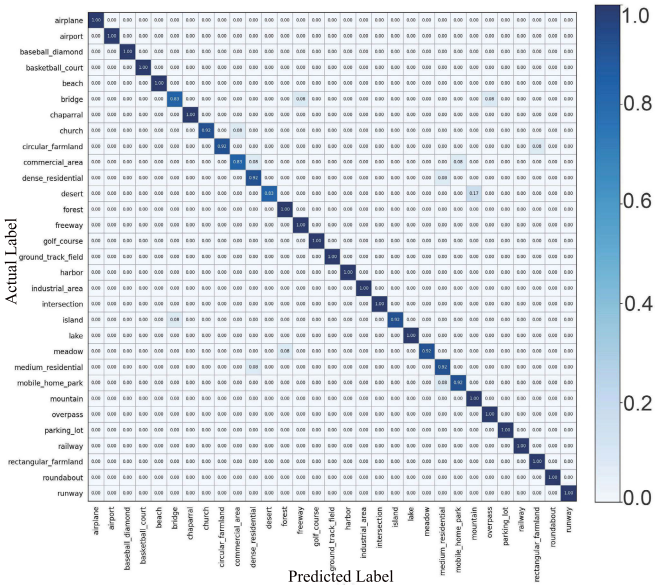


Fig. 8. CM on OPTIMAL-31 when the proportion of trainable data is 80% and OA equals 97.31%.

(over KFBNet(VGGNet-16) [18]), 1.75 (over KFBNet(DenseNet-121) [18]), 1.38 (over ViT-B [13]), 1.33 (over ACGLNet [33]), 0.4 (over VSDNet-DenseNet121 [49]), and 0.85 (over MBLANet [6]), These promising results demonstrate the superiority of our method for classifying RS scenes more accurately.

The CM of our CGINet on OPTIMAL-31 with 80% of the trainable data is displayed in Fig. 8. It can be observed that the accuracy of CGINet exceeds 90% in 29 of the 31 categories. The experimental results once again demonstrate the excellent performance of our method on OPTIMAL-31, which is also demonstrated by the high Kappa (0.9639) in Table V. However, CGINet was confused in distinguishing certain scene categories, such as "Bridge" and "Desert," because these scenes have extremely similar geographical features.

D. Ablation Study

1) Number of Parameters and Computation Complexity: Table VI shows the computational complexity and parameters for one batch on the AID dataset. To obtain more discriminative feature representation, CGINet introduces an LCAB to model the global context and a CELM to learn and locate local discriminative regions of features. Therefore, CGINet has a slight increase in computational complexity compared to the baseline. The increase in computational complexity and parameters of our method is not significant compared to other RS scene classification methods.

TABLE V
KAPPA VALUES OF CGINET FOR THE UCM, AID, NWPU, AND OPTIMAL-31 DATASETS

UCM	AID		NWPU		OPTIMAL-31
80% Training	20% Training	50% Training	10% Training	20% Training	80% Training
0.9962	0.9510	0.9676	0.9193	0.9420	0.9639

TABLE VI
PARAMETERS AND FLOPS OF METHODS ON THE AID DATASET

Methods	Parameters	FLOPs	OA (20%)
GooLeNet [12]	6.04M	1.582G	83.44±0.40
VGGNet16 [34]	134.38M	15.47G	86.59±0.29
ViT-B [13]	85.63M	32.20G	93.81±0.21
Cycle MLP [47]	63.96M	-	95.31±0.15
ResNet50+CBAM [35]	26.10M	4.138G	94.46±0.14
MBLANet [6]	26.10M	4.145G	94.62±0.19
Simplified EMSCNet (ResNet50) [43]	27.49M	-	95.13±0.10
ResNet50 [8]	23.57M	4.132G	94.47±0.18
CGINet (ours)	26.10M	4.140G	95.35±0.14

TABLE VII
COMPARISON OF DIFFERENT CLASSIFICATION LOSSES

Methods	UCM (80%)	OPTIMAL-31 (80%)	AID (50%)	NWPU (10%)
MC Loss [27]	99.52	96.23	96.26	92.08
CC Loss [28]	99.04	95.89	96.64	91.72
CE Loss	99.22	95.96	96.20	91.69
CELM	99.84	97.35	97.10	92.28

2) Evaluate the Statistical Significance of the Classifier: Tests for statistical significance are used to compare whether there are significant differences between the baseline ResNet50 and the proposed CGINet. We chose the paired samples t-test to conduct experiments on UCM and OPTIMAL-31 datasets. First, a null hypothesis is defined: there is no significant difference between CGINet and ResNet50. Next, we initialized the two models with the same random seeds, then changed the seeds and repeated the experiment five times independently on the two datasets. The OA values output by CGINet and ResNet50 are [97.31, 97.04, 97.21, 97.3, 97.28, 99.76, 99.57, 99.8, 99.84, 99.90] and [95.96, 95.95, 95.9, 96, 96.1, 99.25, 99.22, 99.3, 99.23, 99.2], respectively. The probability $p = 2.35062e^{-05}$ was obtained through a paired t-test, which was significantly less than 0.05. Therefore, we reject the null hypothesis and conclude that there is a significant difference between CGINet and ResNet50.

3) Comparison of Different Classification Losses: Table VII shows the comparison between the proposed CELM and other widely used channel losses on four datasets. In all comparison experiments, LCAB is always kept because our final goal is to use both parts together. For the hyperparameters used for comparison, we adaptively tuned the parameters and presented their best results. It can be observed from Table VII that CELM

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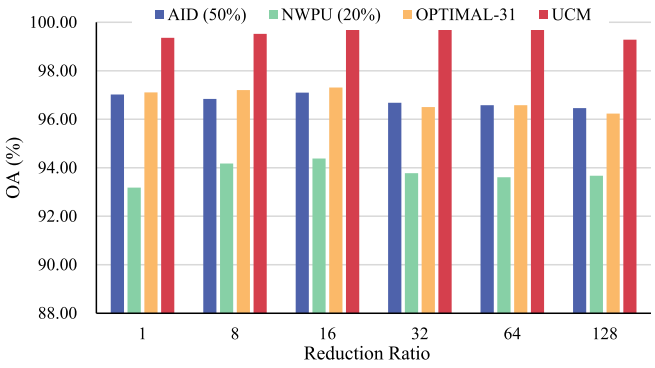


Fig. 9. Effect of the reduction ratio r on different datasets.

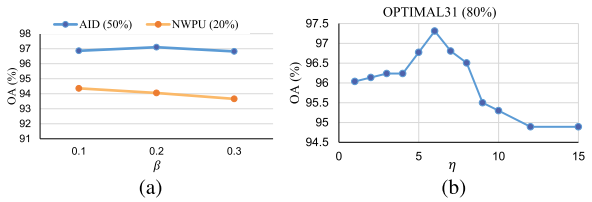


Fig. 10. Effect of hyperparameters in (11) for CELM. (a) Effect of β. (b) Effect of η.

TABLE VIII
ABLATION STUDY OF CELM ON THREE RS SCENE DATASETS

Methods	OPTIMAL-31 (80%)	AID (50%)	NWPU (20%)
ResNet50	95.96	96.20	93.79
LCAB	96.77	96.68	93.97
LCAB + ℓ_{aln}	96.10	96.58	94.07
LCAB + ℓ_{div}	94.62	95.80	93.54
CGINet	97.31	97.10	94.36

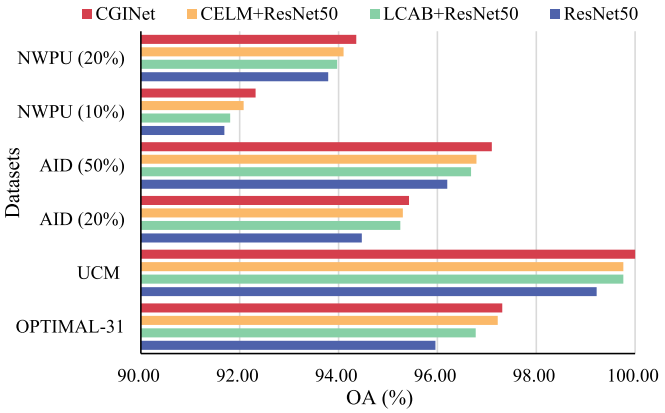


Fig. 11. Effect of each branch in CGINet.

achieves better accuracy. This proves that CELM can more effectively guide the backbone network to distinguish different RS scenes.

4) Ablation Study of Reduction Ratio r: We verify the effect of different reduction ratios r on the experimental results. The reduction rate is directly related to the number of channels in LCAB, which can control the capacity and overhead of the module to a certain extent. As shown in Fig. 9, we compare performance with six different reduction ratios. The best accuracy is achieved with a reduction ratio of 16, even with higher capacities for reduction ratios of 4 and 8. We believe that this phenomenon is caused by the model getting stuck in overfitting. Based on the experimental results, the reduction ratio r is finally set to 16 in our model.

5) Sensitivity Analysis of Parameters in CELM: As shown in Fig. 10, we evaluated the performance of different β and η in (11). β is a controlling hyperparameter to decide the importance of CELM. η indicates the influence of the intragroup diversity part of CELM, while $\beta = 0.1, \eta = 5$ on the NWPU dataset with a training ratio of 20%, and $\beta = 0.2, \eta = 5$ on the AID dataset with a training ratio of 50%. Further increasing the value of β will lead to a performance drop since the model requires the class-label relations provided by CE Loss. As shown in Fig. 10(b), the best performance is obtained when $\beta = 0.3, \eta = 6$ on the OPTIMAL-31 dataset with a training ratio of 80%. The two parts of CELM work together to correctly guide the model to focus on local discriminative regions. If the value of η is too large or too small, it will lead to imbalance and significantly weaken the model performance.

As shown in Table VIII, if we use only the intergroup alignment loss of CELM, the classification accuracies drop by 1.21, 0.52, and 0.32 percentage points on the OPTIMAL-31, AID, and NWPU datasets, respectively. If we use only the within-group diversity loss of CELM, the classification accuracies drop by 2.69, 1.30, and 0.82 percentage points on the OPTIMAL-31, AID, and NWPU datasets, respectively. In CELM, all channel groups influence each other during training. If only one channel group is considered during training, other channel groups may lose diversity. If only the interclass differences are considered during training, the features of the same channel group may converge, degrading model performance.

6) Effectiveness of Each Part: To verify the effectiveness of the individual components of the proposed CGINet, ablation experiments are performed on each part. To be specific, we still choose the pretrained ResNet50 as the backbone and add each part separately. The quantitative comparisons are listed in Fig. 11. We can observe that LCAB outperforms the original ResNet50, which shows the importance of LCAB.

E. Visualization

Discriminative features aim to reflect the differences between different scenes and are directly related to semantic labels. Therefore, the strength of the model can be more intuitively demonstrated by locating the salient regions of the input image. In this article, we employ the Grad-CAM++ [50] to generate energy maps for images from the UCM and AID test sets. Specifically, Grad-CAM++ optimizes the results of Grad-CAM for more accurate positioning. It is more suitable for situations where there is more than one object of the target class in the image. Fig. 12 shows the visualization results of our baseline (ResNet50) and CGINet. As shown in Fig. 12, CGINet classifies "Bridge," "Harbor," and "Basketball court" with salient geospatial objects such as bridges, boats, and courts. This demonstrates that our proposed model can summarize intraclass similarity knowledge. Both the "Harbor" and "Beach" categories contain the ocean. CGINet distinguishes them by boat and land. Both the "Mountain" and "Circular

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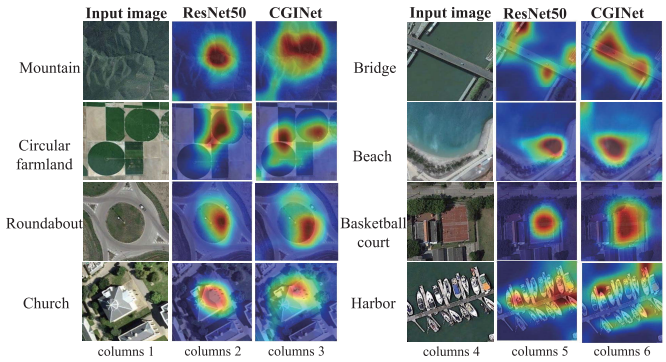


Fig. 12. Grad-CAM++ [50] visualization results. Input images (columns 1 and 4), visualization results from ResNet50 (columns 2 and 5), and visualizing results from the proposed CGINet (columns 3 and 6).

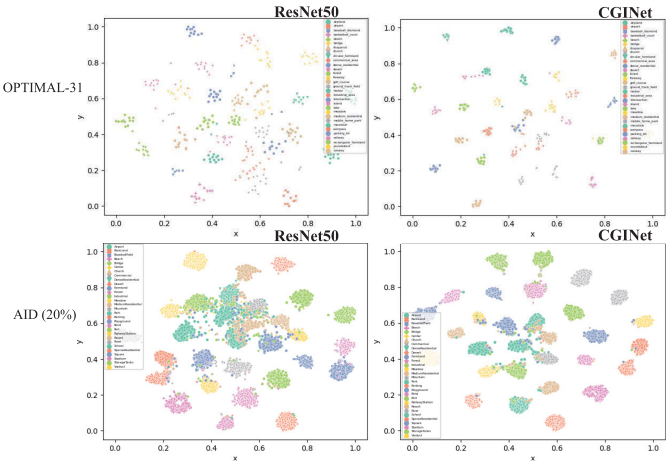


Fig. 13. Comparison of visualization using t-SNE. (Left) Results of the baseline network (pretrained ResNet50) on the test set. (Right) Results of CGINet on the test set.

farmland" categories are dominated by green, and CGINet expands the field of vision for overall understanding, which clearly shows the effectiveness of CGINet in modeling global context. The results of the visualization experiments show that the proposed CGINet can capture discriminative information among different scenes. In addition, it can be seen from the visualization results of ResNet50 that it does have shortcomings in locating local discriminative regions and modeling global context.

In addition, t-distributed stochastic neighbor embedding (t-SNE) [51] is employed to visualize the classification results of ResNet50 and CGINet on the UCM and OPTIMAL-31 test sets. T-SNE maps data from high dimensions to low dimensions and is highly suitable for visualization of high-dimensional data. Specifically, it uses smaller distances to generate larger gradients so that dissimilar data repel each other. As illustrated in Fig. 13, compared with ResNet50, the feature representation generated by CGINet exhibits more obvious characteristics of compactness within classes and dispersion between classes. These findings affirm that CGINet is capable of extracting more discriminative features by employing LCAB to explicitly model global information and CELM to locate the discriminative regions of the feature. Consequently, this provides an intuitive demonstration of the effectiveness of our CGINet.

V. Conclusion

In this article, we propose to combine global context and part-level discriminative features within a unified framework called CGINet for accurate RS scene classification. To this end, we develop an LCAB to explicitly model the global context to obtain larger receptive fields and more contextual information. A CELM is also devised to encourage the model to actively locate discriminative parts for feature enhancement. In CELM, the feature channels are first grouped, and each group represents a class. Then, we directly constrain the feature channels based on two optimization objectives. One objective is to align scene classes and subfeature groups, thus ensuring that each subfeature group only locates the discriminative part features of the target class. The other is to enrich discriminative part features within each subfeature group, thus ensuring that as many local discriminative features as possible can be learned. In particular, CELM is only used during training and not activated during inference, which introduces less computational cost. Moreover, a self-enhancing submodule is proposed to improve the spatial distribution of semantic subfeatures within the group. This enables us to further align class information and feature groups to better fit RS scene images. Comprehensive experiments on four benchmark datasets show consistent improvements achieved by the proposed CGINet.

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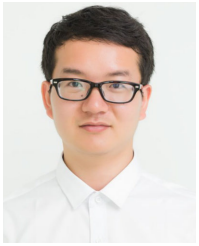
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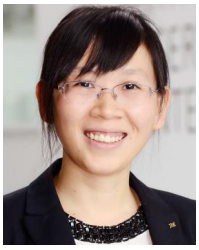


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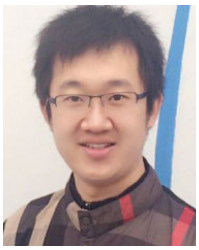
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